Module 6: MLOPS & PRODUCTION DATA SCIENCE

Course Code: 22DS102006 - Model Deployment, Pipelines, Monitoring & AI Ethics

1. Model Packaging & Microservices

Serializing trained estimators (Joblib, ONNX, TorchScript). Building production RESTful inference endpoints with FastAPI, request validation with Pydantic, and async batching.

2. Containerization & CI/CD Pipelines

Docker containerization for reproducible ML environments. Automated testing, linting, model regression benchmarks, and continuous deployment via GitHub Actions and cloud registries.

3. Monitoring, Drift Detection & AI Governance

Tracking concept drift, covariate shift, and latency degradation in live traffic using tools like Evidently AI and Prometheus. Algorithmic fairness audits, explainability with SHAP, and compliance frameworks.